PROJECT: CUSTOMER CHURN PREDICTION

Introduction:

Customer churn prediction is important for companies that provide subscription services like telecom companies. Customer churn means when a customer stops using a company's service.

Losing customers can:

- Reduce revenue
- Decrease market share

Machine learning helps predict which customers may leave. Companies can then take actions such as:

- Offering discounts
- Improving customer service
- Giving loyalty rewards

This helps businesses keep customers and reduce losses**.**

Dataset Description:

The dataset used in this project is the Telecom Customer Churn Dataset available on Kaggle.

Dataset Source:
https://www.kaggle.com/datasets/shilongzhuang/telecom-customer-churn-by-maven-analytics

Dataset name:
Telecom Customer Churn by Maven Analytics Kaggle dataset

The dataset contains customer information related to telecom services, including demographics, service usage, billing details, and subscription information.

	Gender	Age	Married	Number of Dependents	Number of Referrals	Tenure in Months	Offer	Phone Service	Avg Monthly Long Distance Charges	Multiple Lines	...	Contract	Paperless Billing	Payment Method	Monthly Charge	Total Charges	Total Refunds	Total Extra Data Charges	Total Long Distance Charges	Total Revenue	C
0	Female	37	Yes	0	2	9	NaN	Yes	42.39	No	...	One Year	Yes	Credit Card	65.6	593.30	0.00	0	381.51	974.81	
1	Male	46	No	0	0	9	NaN	Yes	10.69	Yes	...	Month-to-Month	No	Credit Card	-4.0	542.40	38.33	10	96.21	610.28	
2	Male	50	No	0	0	4	Offer E	Yes	33.65	No	...	Month-to-Month	Yes	Bank Withdrawal	73.9	280.85	0.00	0	134.60	415.45	
3	Male	78	Yes	0	1	13	Offer D	Yes	27.82	No	...	Month-to-Month	Yes	Bank Withdrawal	98.0	1237.85	0.00	0	361.66	1599.51	
4	Female	75	Yes	0	3	3	NaN	Yes	7.38	No	...	Month-to-Month	Yes	Credit Card	83.9	267.40	0.00	0	22.14	289.54	

5 rows × 31 columns

Important Features in the Dataset:

The dataset contains multiple types of customer attributes:

	Gender	Age	Married	Number of Dependents	Number of Referrals	Tenure in Months	Offer	Phone Service	Avg Monthly Long Distance Charges	Multiple Lines	...	Contract	Paperless Billing	Payment Method	Monthly Charge	Total Charges	Total Refunds	Total Extra Data Charges	Total Long Distance Charges	Total Revenue	C
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Target Variable:

The target variable used for prediction is:

Customer Status
Stayed
Stayed
Churned
Churned
Churned

Data Exploration:

Initial data exploration was performed to understand the dataset structure.

Key checks included:

- Dataset size

```
#show 7043 customers and 37 features.
```

```
(7043, 37)
```

- Missing values

Number of Referrals	0
Tenure in Months	0
Offer	3877
Phone Service	0
Avg Monthly Long Distance Charges	682
Multiple Lines	682
Internet Service	0
Internet Type	1526

- Distribution of churned vs retained customers

```
#show the amound of enteries of  (stayed/joined = 0) and (churned = 1)
```

Customer Status	count
0	5174
1	1869

This indicates that the dataset is slightly **imbalanced,** with more customers staying than leaving.

Data Preprocessing:

Several preprocessing steps were applied to prepare the data for machine learning.

Handling Missing Values:

Missing values in numeric columns were replaced with the **mean value** of the column.
Missing values in categorical columns were replaced with the **most frequent value (mode)**.

```
# Fill numerical columns with mean
df[num_cols] = df[num_cols].fillna(df[num_cols].mean())

# Fill categorical columns with mode
df[cat_cols] = df[cat_cols].fillna(df[cat_cols].mode().iloc[0])
```

Verifying No Missing Value:

Avg Monthly Long Distance Charges	0
Multiple Lines	0
Internet Service	0
Internet Type	0
Avg Monthly GB Download	0
Online Security	0
Online Backup	0
Device Protection Plan	0
Premium Tech Support	0
Streaming TV	0
Streaming Movies	0
Streaming Music	0
Unlimited Data	0

Converting Target Variable:

The target column **Customer Status** was converted into numeric values because machine learning models cannot process text labels.

	Customer Status
0	0
1	0
2	1
3	1
4	1

Encoding Categorical Variables:

Categorical variables such as:

- Gender
- Contract
- Payment Method
- Internet Type

We converted them into numeric format using **One-Hot Encoding**.

```
#encoded categorical columns:

df = pd.get_dummies(df, drop_first=True)

#converted columnes like Gender (male/female etc) into numeric featuers as ML models cannot read text

#Ex:
# 0 = Female
# 1 = Male
```

Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

Shows **Training Images** Split:

```
(5634, 36)
```

Machine Learning Model:

The model used for this project is the **Decision Tree Classifier**. Decision Trees work by splitting the dataset into smaller subsets based on feature values. Each split is selected to best separate the target classes.

Example: rule generated by the model:

- IF Monthly Charge > 80
- AND Contract = Month-to-Month
- THEN Customer Likely to Churn

```
y_pred = model.predict(X_test)
y_pred[:10]

# 0 = stayed
# 1 = chruned
```

```
array([0, 0, 0, 0, 0, 1, 1, 0, 0, 0])
```

Model Evaluation:

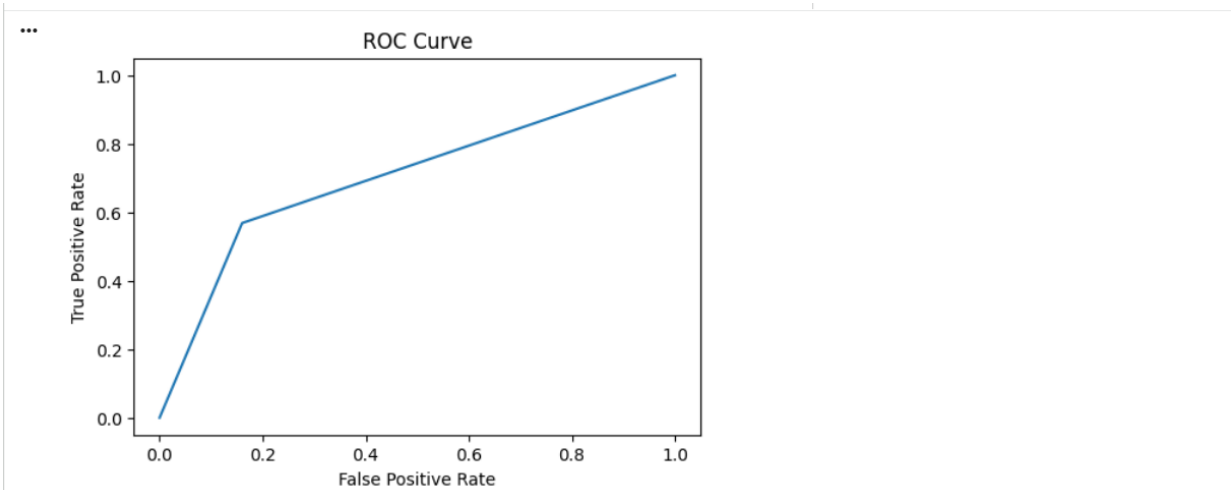
The model was evaluated using several classification metrics.

```
•••  Accuracy: 0.7679205110007097
     Precision: 0.5608465608465608
     Recall: 0.5683646112600537
     F1 Score: 0.5645805592543276
```

```
#Out of all customers, the model correctly predicted churn or stay 76.8%
of the time.
#recall = how many churners were detected
#precision = predicted churners that were correct
#f1 score = balance of prediction and recall
```

ROC Curve:

The **ROC (Receiver Operating Characteristic) Curve** is used to evaluate how well the model distinguishes between churned and non-churned customers.



- The model uses a default probability threshold of 0.5 to classify customers as churned or retained.
- Customers with a predicted churn probability greater than 0.5 are classified as churners,
- While those below the threshold are classified as retained customers.

ROC-AUC Score

The ROC-AUC score measures the **area under the ROC curve**, which represents the overall classification performance.

My model produced:

```
ROC AUC Score: 0.7040664755141968
```

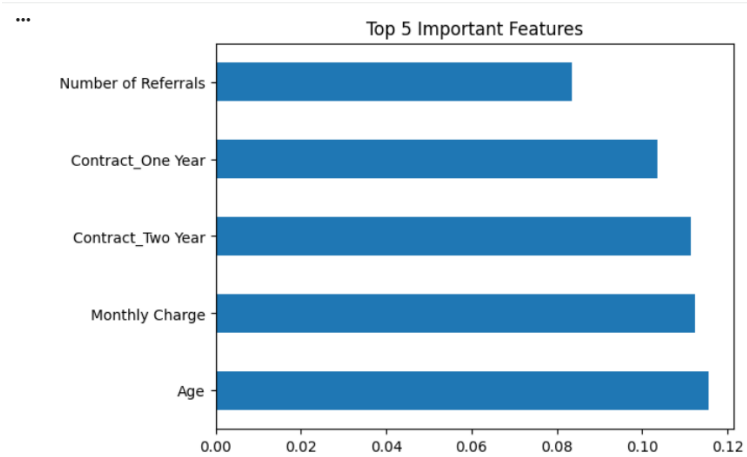
0.70 indicates that the model has a good ability to distinguish between churned and retained customers**.**

Feature Importance Analysis:

Decision Trees allow us to determine which features most influence the prediction.

Top important features identified by the model:

```
Age                    0.115536
Monthly Charge         0.112429
Contract_Two Year      0.111297
Contract_One Year      0.103635
Number of Referrals    0.083383
dtype: float64
```



SHOWS:

Higher value = more influence.
Example: Higher bills often lead to churn

Business Insights:

The analysis reveals several factors that influence customer churn.

Customers Most Likely to Churn

Customers are more likely to churn when they:

- Have high monthly charges
- Use short-term contracts
- Fall into certain age groups

Key Drivers of Churn

The most important churn drivers identified by the model include:

- Contract type
- Monthly charges
- Age
- Number of referrals

Recommended Retention Strategies:

Telecom companies can reduce churn by implementing the following strategies:

1. Offer discounts for long-term contracts.
2. Provide loyalty rewards to customers who refer others.
3. Reduce monthly charges for high-risk customers.
4. Improve service quality and customer support.

These strategies can help businesses retain valuable customers and improve overall profitability.

Conclusion:

In this project I used machine learning to predict customer churn in a telecom company. I trained a Decision Tree model using telecom customer data and it got about 77% accuracy with an ROC-AUC score of 0.70.

The results showed some important reasons why customers leave, like contract type, monthly charges, age, and referrals. These insights can help telecom companies understand their customers better and try to keep them from leaving.

Overall, this project shows that machine learning can help businesses make better decisions and improve customer retention.